

PHYS-467 Machine Learning for Physicists
Exercise session 4: Back to Linear Regression

Exercise n. 1

1. Write a function generating the synthetic dataset $\mathcal{D} = \{x^\mu, y^\mu\}_{\mu=1}^n$, with $x^\mu \sim \mathcal{N}(0, 1)$ and $y^\mu = w_\star \mathbf{x}^\mu + \mathcal{N}(0, \sigma^2)$ with $w_\star \in \mathbb{R}$, then plot it;
2. Infer the parameter $w_\star$ by linear regression;
3. Verify that the maximum of the log-likelihood coincides with the least square solution.

Exercise n. 2

1. Load the diabetes dataset from scikit-learn;
2. Split the dataset in train, test and validation set;
3. Compute the estimator with L_2 regularization and plot its components as a function of the regularization strength;
4. Determine the optimal value of the regularization strength using the validation set;
5. Compute the test error and plot the test labels versus the predicted ones.

Exercise n. 3

1. Repeat points from 3 to 5 of Exercise n. 2 with L_1 regularization. Which are the differences between L_1 and L_2 ?